

Assessment Tool-3

Q1) R program:

Install packages (Run only once)

```
install.packages("ggplot2")
```

```
install.packages("scales")
```

Load libraries

```
library(ggplot2)
```

```
library(scales)
```

Dataset

```
retail <- data.frame(
```

```
  Branch=c("B1","B2","B3","B4","B5","B6","B7","B8","  
B9","B10",
```

```
  "B11","B12","B13","B14","B15","B16","B17","B18","  
B19","B20"),
```

Region=c("North","South","East","West","North","South",
"East","West",
"North","South","East","West","North","South","East",
"West",
"North","South","East","West"),

Category=c("Electronics","Furniture","Clothing","Food",
"Electronics",
"Furniture","Clothing","Food","Electronics","Furniture",
",
"Clothing","Food","Electronics","Furniture","Clothing",
,
"Food","Electronics","Furniture","Clothing","Food"),

Sales=c(450000,620000,180000,920000,350000,72000
0,250000,1080000,
560000,670000,195000,980000,430000,760000,280000
,1150000,
510000,690000,220000,1250000),

Profit=c(45000,95000,-
12000,185000,25000,120000,18000,225000,

```
72000,98000,-  
8000,205000,51000,132000,24000,250000,  
69000,115000,15000,285000),
```

```
Customers=c(1200,2500,980,4200,1800,3000,1250,520  
0,  
2100,3300,1100,4700,1750,3600,1450,5600,  
2200,3900,1600,6100)  
)
```

```
# Visualization
```

```
ggplot(retail,  
  aes(x = Sales,  
      y = Profit,  
      size = Customers,  
      colour = Category,  
      shape = Region)) +  
  geom_point(alpha = 0.8) +  
  geom_text(aes(label = Branch),  
    vjust = -0.7,  
    size = 3) +
```

```

scale_x_log10(labels = comma) +
scale_size(range = c(4,12)) +
labs(
  title = "Retail Sales Performance Across Branches",
  subtitle = "Bubble Chart with Logarithmic Sales
Scale",
  x = "Sales (Log Scale)",
  y = "Profit",
  colour = "Product Category",
  shape = "Region",
  size = "Customers"
) +
theme_minimal() +
theme(
  plot.title = element_text(size=16, face="bold"),
  legend.position = "right"
)

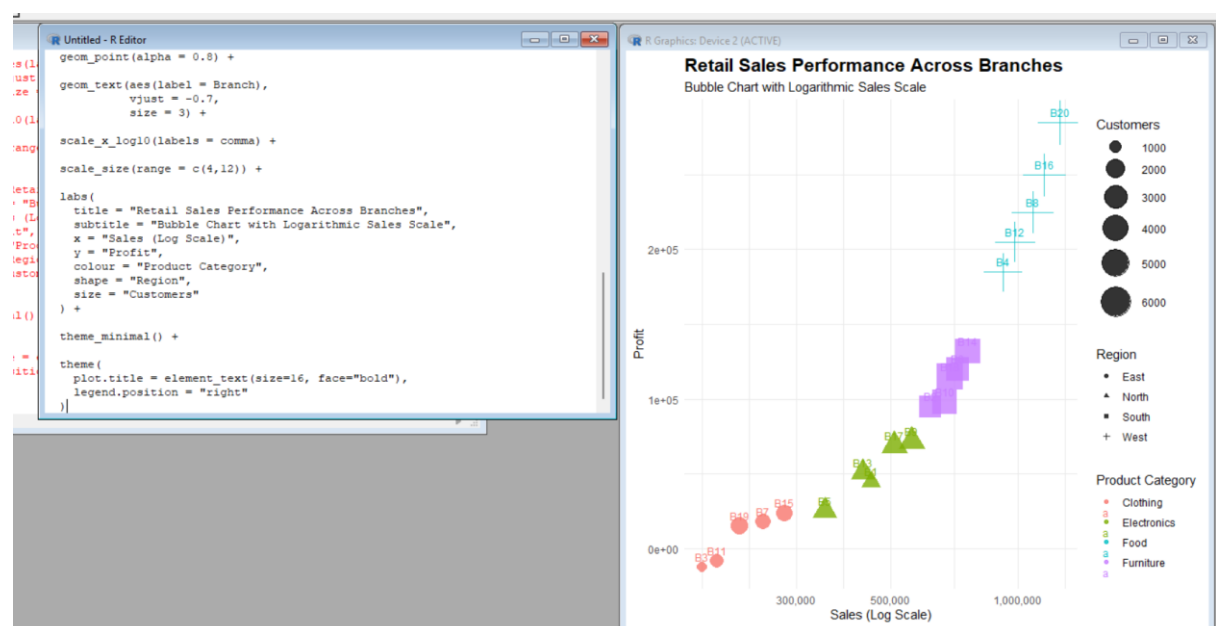
```

Justification

- A bubble chart is chosen because it can display multiple variables simultaneously.

- Sales is placed on the X-axis and Profit on the Y-axis to study their relationship.
- Bubble size represents the number of customers, allowing easy comparison of customer strength across branches.
- Different colours distinguish product categories.
- Different shapes identify regions.
- Branch labels make each point identifiable.
- A logarithmic X-axis improves readability by handling the wide range of sales values.
- The minimal theme keeps the visualization clean and professional.

Output:



Q2) R program:

```
# Install package (Run only once)
```

```
install.packages("ggplot2")
```

```
install.packages("reshape2")
```

```
# Load libraries
```

```
library(ggplot2)
```

```
library(reshape2)
```

```
# Dataset
```

```
energy <- data.frame(
```

```
  Month = month.abb,
```

```
  Coal =
```

```
c(620,580,590,605,640,670,690,720,700,680,650,630),
```

```
  Gas =
```

```
c(310,295,305,320,340,355,360,370,365,350,335,320),
```

```
  Solar =
```

```
c(45,60,90,130,180,230,250,245,210,160,90,50),
```

```
  Wind =
```

```
c(80,95,120,150,170,185,195,190,175,150,110,90),
```

```
Hydro =  
c(140,150,160,170,180,195,205,210,200,185,165,150)  
)
```

```
# Convert data to long format  
energy_long <- melt(energy, id.vars = "Month",  
                    variable.name = "Source",  
                    value.name = "Generation")
```

```
#-----  
# Cartesian Coordinate Visualization  
#-----
```

```
ggplot(energy_long,  
       aes(x = Month,  
           y = Generation,  
           colour = Source,  
           group = Source)) +  
geom_line(size = 1.2) +  
geom_point(size = 2.5) +
```

```
labs(  
  title = "Monthly Electricity Generation (Cartesian  
Coordinates)",  
  x = "Month",  
  y = "Generation (MW)",  
  colour = "Energy Source"  
) +  
theme_minimal()
```

```
#-----  
# Polar Coordinate Visualization  
#-----
```

```
ggplot(energy_long,  
  aes(x = Month,  
    y = Generation,  
    fill = Source)) +  
geom_col(position = "stack") +  
coord_polar() +  
labs(
```



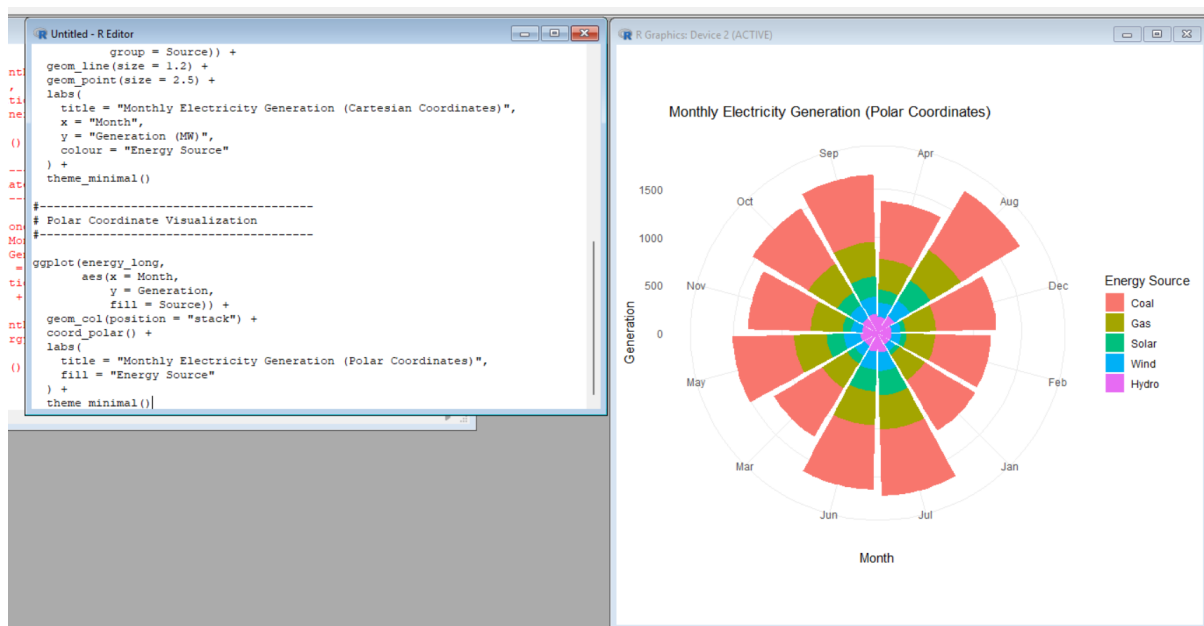
```
title = "Monthly Electricity Generation (Polar  
Coordinates)",  
fill = "Energy Source"  
) +  
theme_minimal()
```

Justification :

- **Cartesian Coordinate System:** A line chart is chosen because it clearly shows monthly trends and allows easy comparison of electricity generation among different energy sources. Different colours help distinguish each source, while points highlight the monthly values.
- **Polar Coordinate System:** A polar chart is used to represent the data in a circular form, making it suitable for displaying cyclical patterns across the 12 months. Different colours improve the identification of each energy source.
- **Comparison:** The Cartesian coordinate system provides better analytical insight because it enables accurate comparison of values, trends, and seasonal variations. Although the polar coordinate system is visually appealing and effectively represents the cyclical nature of monthly data, it makes precise value comparison more difficult.

- Conclusion: Therefore, the Cartesian coordinate system is the preferred choice for data analysis, while the Polar coordinate system is more suitable for presentation and visual exploration.

Output:



Q3) R program:

Install packages (Run only once)

```
install.packages("ggplot2")
```

```
install.packages("viridis")
```

Load libraries

```
library(ggplot2)
```

```
library(viridis)
```

```
# Create dataset

climate <- data.frame(

City=c("Delhi","Mumbai","Chennai","Kolkata","Hyder
abad",

"Bangalore","Pune","Ahmedabad","Jaipur","Lucknow",

"Bhopal","Patna","Goa","Kochi","Nagpur","Indore",

"Surat","Visakhapatnam","Coimbatore","Mysuru"),

Temperature=c(44,35,38,40,39,31,33,43,45,39,

38,40,32,30,41,37,36,34,29,28),

Humidity=c(22,84,78,72,48,66,58,26,18,42,

36,45,89,91,30,34,76,82,68,64),

Rainfall=c(6,280,220,190,85,120,95,2,1,35,

48,72,360,410,12,20,180,240,130,145),

AQI=c(365,110,125,180,160,72,84,290,320,240,

175,220,58,52,270,190,145,98,60,55)
```

)

Bubble Chart

ggplot(climate,

aes(x = Temperature,

y = AQI,

size = Rainfall,

colour = Humidity,

label = City)) +

geom_point(alpha = 0.8) +

geom_text(vjust = -0.8, size = 3) +

scale_colour_viridis_c(option = "plasma") +

scale_size(range = c(3,12)) +

labs(

```
title = "Climate Conditions Across Major Indian  
Cities",  
x = "Temperature (°C)",  
y = "Air Quality Index (AQI)",  
colour = "Humidity (%)",  
size = "Rainfall (mm)"  
) +  
  
theme_minimal()
```

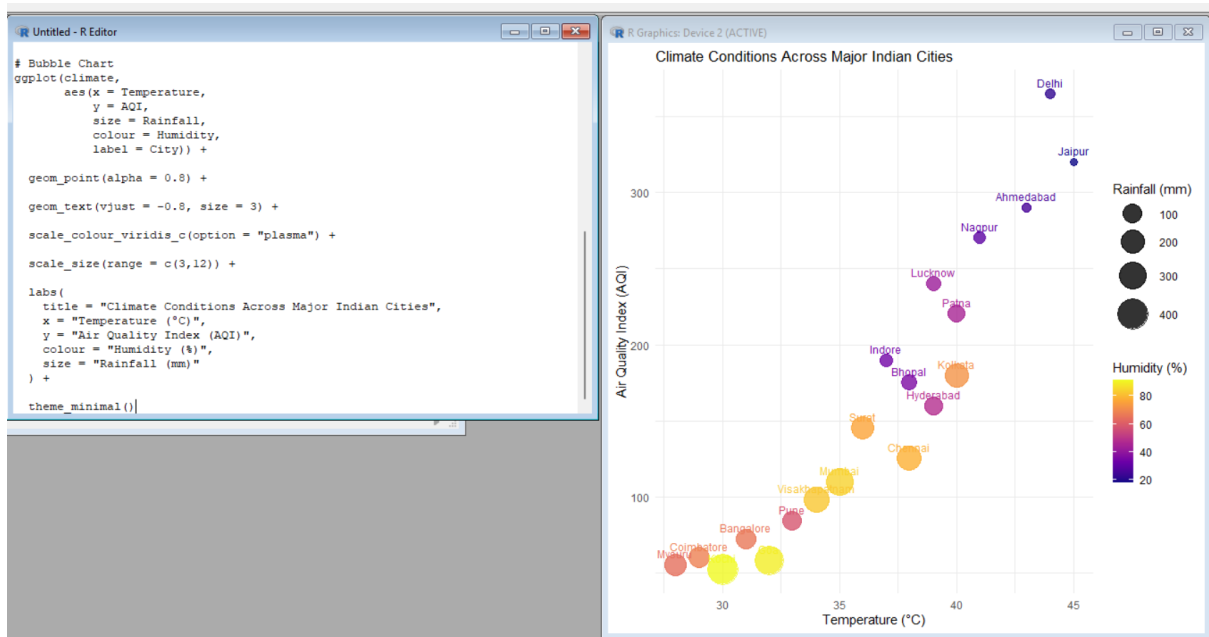
Justification

- A bubble chart is selected because it effectively represents four variables in a single visualization.
- Temperature is mapped to the X-axis and AQI to the Y-axis, making it easy to observe the relationship between temperature and air quality.
- Rainfall is represented using bubble size, allowing cities with higher rainfall to stand out visually.
- Humidity is represented using a continuous colour scale.
- The Viridis ("Plasma") colour palette is used because it is colour-blind friendly, perceptually

uniform, and provides clear distinctions between low and high humidity values.

- City names are displayed as labels to identify each observation.
- The minimal theme reduces visual clutter and improves readability.
- This visualization enables users to compare temperature, AQI, rainfall, and humidity simultaneously while ensuring accessibility and accurate interpretation.

Output:



Q4) R program

Install package (Run only once)

```
install.packages("ggplot2")
```

```
# Load library
```

```
library(ggplot2)
```

```
# Create dataset
```

```
software <- data.frame(  
Team=paste("T",1:20,sep=""),
```

```
LinesOfCode=c(12000,18000,25000,35000,48000,  
62000,81000,96000,120000,150000,  
185000,240000,310000,390000,480000,  
620000,810000,1050000,1350000,1800000),
```

```
Bugs=c(35,52,60,72,88,  
110,125,142,165,188,  
205,240,280,320,365,  
420,490,560,640,720),
```

```
Developers=c(5,7,9,10,12,
```

14,16,18,20,22,
25,28,30,34,38,
42,46,50,55,60),

Complexity=c(2.1,2.4,2.7,3.0,3.2,
3.5,3.8,4.1,4.5,4.8,
5.1,5.4,5.8,6.2,6.6,
7.1,7.6,8.0,8.5,9.0)
)

Scatter Plot with Logarithmic Scale

```
ggplot(software,  
  aes(x = LinesOfCode,  
    y = Bugs,  
    size = Developers,  
    colour = Complexity,  
    label = Team)) +
```

```
geom_point(alpha = 0.8) +
```

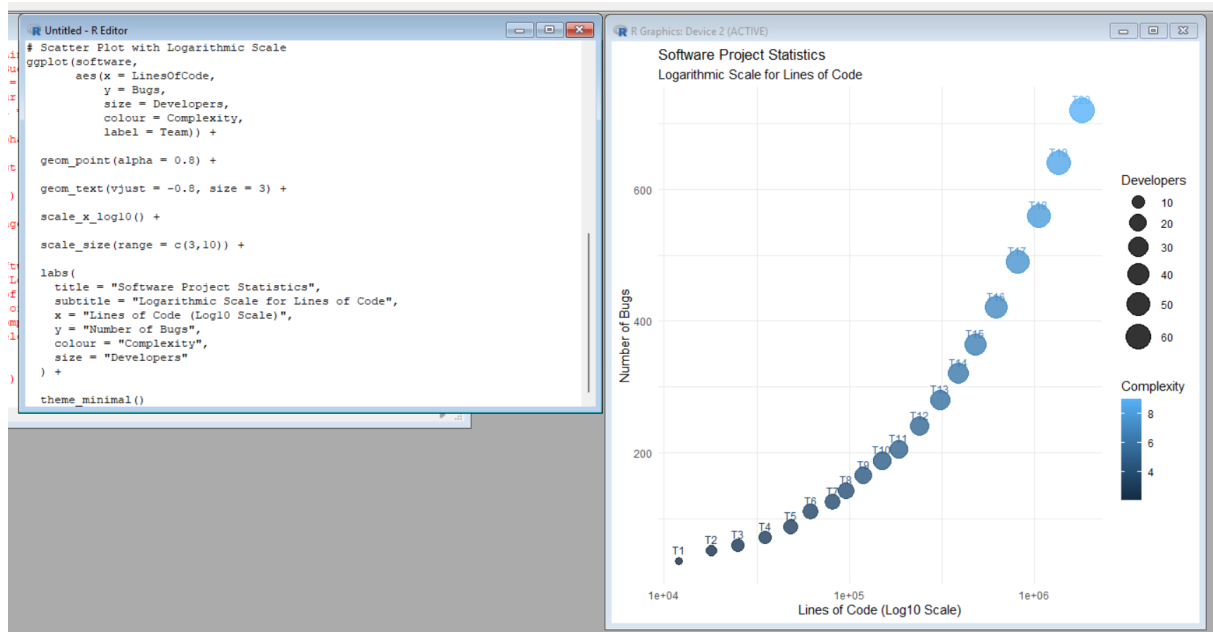


```
geom_text(vjust = -0.8, size = 3) +  
  
scale_x_log10() +  
  
scale_size(range = c(3,10)) +  
  
labs(  
  title = "Software Project Statistics",  
  subtitle = "Logarithmic Scale for Lines of Code",  
  x = "Lines of Code (Log10 Scale)",  
  y = "Number of Bugs",  
  colour = "Complexity",  
  size = "Developers"  
) +  
  
theme_minimal()
```

Justification

- A scatter plot is chosen because it effectively shows the relationship between Lines of Code and Number of Bugs.
- Lines of Code (LOC) is mapped to the X-axis, while Bugs are mapped to the Y-axis.
- Developers are represented by the size of the points, making team size easy to compare.
- Complexity is represented using a colour gradient, highlighting projects with different complexity levels.
- A logarithmic ($\log_{10}$) transformation is applied to the X-axis because the LOC values are highly skewed, ranging from 12,000 to 1,800,000.
- The logarithmic scale compresses large values and spreads smaller values, making all projects visible and easier to compare.
- On a linear scale, smaller projects would cluster together and larger projects would dominate the graph, making interpretation difficult.

Output:



Q5) R program

Load libraries

library(ggplot2)

library(viridis)

Dataset

companies <- data.frame(

Company=paste("C",1:20,sep=""),

Sector=c("IT","Banking","Healthcare","Retail","Energy",

"IT","Banking","Healthcare","Retail","Energy",

"IT","Banking","Healthcare","Retail","Energy",
 "IT","Banking","Healthcare","Retail","Energy"),
 Revenue=c(420,580,720,850,980,
 1150,1320,1480,1650,1820,
 2050,2280,2550,2840,3150,
 3480,3860,4250,4680,5200),
 ProfitMargin=c(8.2,15.6,11.4,9.3,18.5,
 20.4,17.8,14.2,10.5,22.1,
 24.5,19.8,16.7,13.4,26.2,
 28.1,21.5,18.3,15.2,30.4),
 Employees=c(1500,3200,2100,4200,2800,
 5100,6800,3900,7200,4500,
 8100,9500,6200,10300,7200,
 11800,13200,8600,14500,16000),
 MarketCap=c(8500,12000,9800,7600,14500,
 18200,21500,16800,13200,24500,
 28500,32500,24800,19200,38200,
 42500,46800,35600,29800,52000)
)

```
# Integrated Visualization
ggplot(companies,
       aes(x = Revenue,
           y = ProfitMargin,
           size = MarketCap,
           colour = Sector)) +

  geom_point(alpha = 0.8) +

  geom_text(aes(label = Company),
            vjust = -0.7,
            size = 3) +

  scale_size_continuous(range = c(4,14),
                        name = "Market Capitalization") +

  scale_colour_viridis_d(option = "D") +

  labs(
    title = "Financial Performance of Public Companies",
```

```
x = "Revenue (Million)",  
y = "Profit Margin (%)",  
colour = "Industry Sector"  
) +
```

```
theme_minimal(base_size = 14)
```

Justification

- A bubble chart is chosen because it effectively represents multiple variables in a single visualization and allows easy comparison between companies.
- Revenue is mapped to the X-axis, while Profit Margin is mapped to the Y-axis to clearly show the relationship between company revenue and profitability.
- Market Capitalization is represented by the size of the bubbles, making it easy to compare the market value of different companies.
- Industry Sector is represented using different colours, allowing companies from various sectors to be easily distinguished.

- Company names are displayed as labels to identify each company without referring to an external table.
- A Cartesian coordinate system is used because it accurately represents continuous numerical variables and makes trends and comparisons easy to interpret.
- A continuous size scale is applied to the bubbles so that Market Capitalization is represented proportionally without making bubbles too small or too large.
- The Viridis colour palette is used because it is colour-blind friendly, perceptually uniform, and reduces perceptual bias while providing clear distinction between industry sectors.

Output:

